

Skills Summary

Multiplying by a One-Digit Number

Use this reference while completing your worksheet or when studying.

The big idea. Multiplying by one digit means multiplying that digit by *every* place of the other number — ones, tens, hundreds, thousands — and then putting the places back together.

The regrouping rule: multiply first, *then* add the digit you carried in. If the result is ten or more, write its ones digit and carry the tens digit into the next column.

A zero is a place, not a gap. A zero digit still gets multiplied, and it still receives any digit carried into it.

1. Multiply a big number by one digit

Work right to left: ones, tens, hundreds, thousands. Write the carry above the next column so you cannot forget it, and cross it off once you have added it.

Example: Multiply 1362×4 .

Step 1. Every place of 1362 has to be multiplied by the same factor, so work right to left and carry when a column reaches ten.

Step 2. Ones: $2 \times 4 = 8$. Tens: $6 \times 4 = 24$, write 4 and carry 2. Hundreds: $3 \times 4 = 12$, plus the carried 2 is 14 — write 4, carry 1. Thousands: $1 \times 4 = 4$, plus 1 is 5.

The product is **5448**.

Try it: Multiply 2438×3 .

check: 7314

2. Estimate a product by rounding first

Round the big factor to a friendly place, keep the one-digit factor as it is, then multiply. An estimate tells you what size the real answer should be, so a product far from it is a signal to check your work.

Example: Estimate 3847×6 , then find the exact product.

Step 1. The question asks how big the answer should be, so round the four-digit factor to a place that is easy to multiply and leave the 6 alone.

Step 2. To the nearest thousand 3847 becomes 4000, and 4000×6 is 24 thousands.

Estimate: **24000**. Exact: **23082**, only a little below it.

Try it: Estimate 5216×7 by rounding to the nearest thousand, then find the exact product.

check: estimate 36512, exact 35000

3. Multiply to answer a story question

Equal groups mean multiply: *how many groups* times *how many in each group*. Read the last sentence again before you answer — some questions ask for the total, and some ask for what is left after the total.

Example: A theater has 4 sections, and each section holds 1145 seats. How many seats in all?

Step 1. Four sections of the same size are four equal groups, and equal groups are what multiplication counts.

Step 2. Multiply the seats in one section by the number of sections: 1145×4 , which is 4 thousands, 4 hundreds, 8 tens and 0 ones.

There are **4580** seats.

Try it: A farm collects 2260 eggs each day for 8 days. How many eggs is that?

check: 18080

(!) Watch out: the two mistakes that cost the most marks are adding the carry *before* multiplying instead of after, and skipping a zero digit. Both leave a product that is far from your estimate — which is exactly why the estimate is worth writing down first.